



Procedure

- **Initialization:** Prepare a quantum register in a superposition state using Hadamard gates, and another register in the state of the quantum system being studied.
- **Controlled Unitary Operations:** Apply controlled unitary operations that evolve the system's state, encoding the phase information into the first quantum register.
- **Inverse Quantum Fourier Transform (QFT):** Perform an inverse QFT on the first register, translating the encoded phase information into readable outputs.
- **Measurement:** Measure the first register to obtain an estimate of the phase (eigenvalue).

In terms of applications, QPE is pivotal for algorithms that simulate molecular energies, facilitating breakthroughs in drug discovery and material science by predicting molecular properties and reactions.

Variational Quantum Eigensolver (VQE)

VQE addresses the challenge of finding the ground state energy of a molecule, a problem essential for understanding chemical reactions and designing new materials.

Approach

- **Ansatz Preparation:** Choose an ansatz (a guess of the wavefunction) and prepare a quantum circuit that represents it.
- **Parameter Optimization:** Use a classical optimizer to iteratively adjust the parameters of the quantum circuit, minimising the energy expectation value.
- **Measurement and Feedback:** Measure the output of the

circuit, feed the results back to the classical optimizer, and repeat the process until the ground state energy is found.

What is its significance, you may ask? Well, VQE offers a promising method for simulating quantum systems on near-term quantum devices, bridging the gap between current technological capabilities and the requirements for full-scale quantum simulation.

Quantum Machine Learning Algorithms

QML algorithms explore the integration of quantum computing into machine learning, aiming to process and analyse data in fundamentally novel ways that could surpass classical machine learning techniques.

Quantum Approximate Optimization Algorithm (QAOA)

The QAOA is designed to solve combinatorial optimization problems, finding optimal solutions by preparing quantum states that encode potential answers.

Mechanism

- **State Preparation:** Initialise the system in a superposition of all possible solutions.
- **Parameterized Quantum Circuits:** Apply a sequence of quantum gates controlled by parameters that are iteratively optimised.
- **Measurement and Optimization:** Measure the outcome, use classical optimization techniques to adjust the parameters, and repeat the process to converge on the optimal solution.